Justifications of Dyadic Harm in Political Manifestos

Morgan Ballesteros

University of Chicago

Author Note

Correspondence concerning this article should be addressed to Morgan Ballesteros. E-mail: mballesteros@uchicago.edu

Justifications of Dyadic Harm in Political Manifestos

The theory of dyadic morality suggests that judgments of harm give moral valence to relationships between an intentional agent and vulnerable patient (Schein & Gray, 2018). That is, dyadic morality predicts that if one were to see someone (i.e., an agent) intentionally harming another person (i.e., a vulnerable patient), one would judge that this action (i.e., the harming of another person) is immoral or morally wrong. Furthermore, social interactions are composed of moral norms as mutually agreed upon rules as to how citizens ought to behave. Although moral norms (e.g., do not murder) are meant to keep people from damaging others, some justify norm violations — and harm — in pursuit of a goal (e.g., political assassinations), whereas others never condone violence to induce change. Understanding what differentiates violent and non-violent agents is therefore useful in protecting against harmful acts. One source that marks such differences is the political manifesto (i.e., "a written statement declaring publicly the intentions, motives, or views of its issuer"; Merriam-Webster, n.d.). The current study targets differences in linguistic indicators of justifications of harm and integrative complexity between violence-condoning and non-violent manifestos.

Research on linguistic markers of political violence follows the growing number of far-right manifestos published online prior to terrorist attacks (Capelos, DaVisio, & Salmela, 2025; Ebner, Kavanagh, & Whitehouse, 2025). Some of these studies evaluate manifestos through the fusion-plus-threat model of violent extremism (Whitehouse, 2018), which predicts that individuals who lump their personal and group identities together will be more likely to self-sacrifice for their group. Associated linguistic work suggests that words indicating identity fusion, group threat, and violence-condoning norms — including justifications of violence — are correlated with terrorism (Ebner, Kavanagh, & Whitehouse, 2024; Ebner et al., 2025). While the existing literature grounds the importance of justificatory rhetoric in predicting terrorism, harm has not yet been examined as central to moral judgment despite its importance in some theories of moral

judgment (e.g., the theory of dyadic morality). The present study serves as a real-world case on how emotion and reason interact to produce the judgments and justifications necessary for enacting harm. Connecting the political manifesto and moral judgment literatures will enlighten respective research as to (a) how moral reasoning implicates justification in manifestos and (b) how justification operates in the moral reasoning process, each of which could have policy implications.

Integrative Complexity is a linguistic measure of cognitive complexity composed of two primary components: (a) Differentiation and (b) Integration (Suedfeld, Tetlock, & Streufert, 1992). Differentiation indicates how many perspectives one accounts for; Integration indicates the extent to which those differentiated perspectives are synthesized. A useful example is an argumentative essay, which, if it is complex, recognizes a variety of possible arguments for or against some claim, and synthesizes those arguments into one ultimate argument, using parts of each subsidiary argument (Ballesteros, 2025). Integrative complexity shows a robust relationship with political violence (Suedfeld, 2010). However, research using the construct as a predictor of political violence has largely focused on political leaders in respect to events like war and international crises (Suedfeld, 2010). To my knowledge, no study to date has directly linked the integrative complexity of political manifestos with justifications of political violence. Considering the measure's long-standing precedent in predicting political events, integrating integrative complexity into the manifesto literature could help identify the factors that distinguish between merely violent rhetoric and rhetoric that induces violence, thereby affording greater predictability through an aggregation of said factors.

Current Study

The current study contributes to research seeking to identify differences between violent and non-violent political agents through an evaluation of political manifestos in two new ways: (a) by focusing a qualitative analysis on rhetoric about harm as designating

moral valence by the theory of dyadic morality and (b) by scoring manifestos on integrative complexity. I seek to aid in answering three related questions: (a) do justifications of harm correspond to changes in individual moral norms?; (b) does integrative complexity predict whether a manifesto will result in violence versus non-violence?; and (c) does integrative complexity predict both temporally near and far instances of violence in relation to political manifestos? I predict that (a) changes in moral norms — as exemplified in the agents' writings — and subsequent justification of increasingly harmful acts will be associated with a higher likelihood of violence, (b) lower levels of integrative complexity in political manifestos will predict higher likelihoods of violence, and (c) a decrease in integrative complexity will be associated with an individual coming to believe that harm is justified in pursuit of their political goals.

Method

Manifesto Selection

This study includes manifestos divided into three categories following Ebner et al. (2025); (a) violent self-sacrificial, (b) ideologically extreme, and (c) moderate. It also affords longitudinal analysis. I selected not only one-off manifestos already analyzed in the aforementioned literature (e.g., Anders Behring Breivik's Manifesto of the Norway Attacks, 2011; Marx & Engels Manifesto of the Communist Party, 1848), but also sets of writings of individual authors across time (e.g., selections of Vladimir Lenin's letters). Materials include ten manifestos for each of the aforementioned categories, and three longitudinal sets of writings for each category. Incorporating the 15 manifestos previously analyzed in the literature brings the study to a categorical count as follows: (a) nine violent self-sacrificial texts, (b) three ideologically extreme texts, and (c) 3 moderate texts. Each of these texts are one-off samples. To fill out the rest of the ten one-off samples for each category, manifestos were selected according to their place on the political spectrum (i.e.,

left to right political platforms), individual actor-author texts versus platform-representative texts, and length in order to match each category along these factors. The same factors were considered in choosing longitudinal samples, and selection included qualitatively matching samples across categories.

Measures

Linguistic Indicators. Ebner et al. (2025) supplies a list of words that indicate justifications of violence and, as a higher-order category of linguistic markers, violence condoning norms. The current study uses the set of words indicating violence condoning norms with minor alterations. First, indicators like "need for jihaad" (Ebner et al., 2025) have been replaced with relevant phrasing (e.g., need for war; need for revolution). Second, specific indicators of talk about harm in dyadic morality (e.g., harm, damage, agent, patient, intentional) have been added to the list of indicators.

Automated Integrative Complexity. Hand-scoring integrative complexity is an involved process requiring multiple scorers and often hours of work per scored material. In light of time and manpower limitations, this study instead uses automated integrative complexity (Conway, Conway, Gornick, & Houck, 2014) to score materials, a measure that uses linguistic markers of differentiation and integration to construct an automated account of integrative complexity. While automated integrative complexity does not correlate perfectly with hand scores (document-level correlation of .82, paragraph-level correlation of .46, Conway, Conway, & Houck, 2020), it is an increasingly validated measure in its own right and often used as an efficient and effective replacement for hand scoring. This study uses the document-level scores for all analyses.

Violence. Previously analyzed manifestos are easily delineated as to whether they cause violence; some manifestos precede lone-actor attacks by their author, the other manifestos did not precede violence enacted by their author. The present study maintains this distinction. However, it also includes platform-based violence by identifying

institutional violence in direct causal relationship with the philosophies put forward by corresponding manifestos. For example, a manifesto by a political leader that accompanies a war (e.g., the *Russian Imperial Manifesto of 1914* by Tzar Nicholas II announcing Russia's entry into World War One; Golder, 1927) would be counted as violent; a manifesto espoused by a political leader that does not accompany institutional violence by that leader's institution (e.g., John F. Kennedy's *Ich bin ein Berliner*) would be counted as non-violent.

Analyses. Following Ebner et al. (2024), I will measure the prevalence of various categories of linguistic markers indicating violence condoning norms (i.e., justification of violence, martyrdom narrative, hopelessness of alternatives, calls to violence) and harm as a percentage of occurrence relative to total sentence count. These prevalence rates will be compared across manifesto categories (i.e., violent versus non-violent, extreme versus non-extreme) to replicate Ebner et al. (2024) results suggesting that justification of violence is a significant predictor between categories. I will create .zip files for each selected manifesto and run them through automated integrative complexity, which will give scores for integrative complexity, differentiation, and integration. These scores will be compared in the same manner as prevalence rates. I will also run regression analyses for longitudinal manifesto sets in order to understand the effect changes in integrative complexity have on likelihood of violence. I will repeat these analyses for linguistic markers of harm on said longitudinal sets.

Results

```
doc_scores <- doc_scores |>
  dplyr::filter(!is.na(category))


nrow(doc_scores)
```

```
## [1] 27
```

```r
table(doc_scores$category, useNA = "ifany")
```

```
##
##  extreme moderate  violent
##       10        9        8
```

```r
table(doc_scores$category, useNA = "ifany")
```

```
##
##  extreme moderate  violent
##       10        9        8
```

```r
nrow(doc_meta)
```

```
## [1] 31
```

```r
nrow(doc_scores)
```

```
## [1] 27
```

```r
head(doc_meta$doc_id, 20)
```

```
##  [1] "1160kennedyichbin"
##  [2] "142thunberg"
##  [3] "1945labourpartymanifesto"
##  [4] "1949simonedebeauvoirthesecondsex"
##  [5] "atlanticcharter"
```

```
##  [6] "communistmanifesto"
##  [7] "defeatingeurabia1"
##  [8] "earnestmanifesto042719"
##  [9] "fourfreedomsspeechfranklindelanoroosevelt1"
## [10] "fulltextihaveadream"
## [11] "hitleradolfmeinkampf"
## [12] "imperiumyockey"
## [13] "maududisayyidabulalaaljihadfilislambysyedabulaalamaududi"
## [14] "piercewilliamlfreespeechessays19952002"
## [15] "porthuronstatement"
## [16] "revoltagainstthemodernworldjuliusevola"
## [17] "russianimperialmanifestoannouncingthestartofwarinaugust1914"
## [18] "andersbreivik2083manifesto2011"
## [19] "brentontarrantthegreatreplacementmanifesto"
## [20] "chomanifesto11"
```

```r
head(doc_scores$doc_id, 20)
```

```
##  [1] "1160kennedyichbin"
##  [2] "142thunberg"
##  [3] "1945labourpartymanifesto"
##  [4] "1949simonedebeauvoirthesecondsex"
##  [5] "atlanticcharter"
##  [6] "communistmanifesto"
##  [7] "defeatingeurabia1"
##  [8] "earnestmanifesto042719"
##  [9] "fourfreedomsspeechfranklindelanoroosevelt1"
## [10] "fulltextihaveadream"
```

```
## [11] "hitleradolfmeinkampf"
## [12] "imperiumyockey"
## [13] "maududisayyidabulalaaljihadfilislambysyedabulaalamaududi"
## [14] "porthuronstatement"
## [15] "revoltagainstthemodernworldjuliusevola"
## [16] "russianimperialmanifestoannouncingthestartofwarinaugust1914"
## [17] "brentontarrantthegreatreplacementmanifesto"
## [18] "chomanifesto11"
## [19] "milestonesf1"
## [20] "patrickcrusiustheinconvenienttruthaboutme"
```

```r
table(doc_scores$category, useNA = "ifany")
```

```
##
##  extreme moderate  violent
##       10        9        8
```

```r
doc_scores %>%
  dplyr::filter(is.na(category)) %>%
  dplyr::select(doc_id, corpus, author) %>%
  head(20)
```

```
## # A tibble: 0 x 3
## # i 3 variables: doc_id <chr>, corpus <chr>, author <chr>
```

```r
names(doc_meta)
```

```
## [1] "doc_id"   "category"
```

```r
head(doc_meta)
```

```
## # A tibble: 6 x 2
##   doc_id                           category
##   <chr>                            <chr>
## 1 1160kennedyichbin                moderate
## 2 142thunberg                      moderate
## 3 1945labourpartymanifesto         moderate
## 4 1949simonedebeauvoirthesecondsex moderate
## 5 atlanticcharter                  moderate
## 6 communistmanifesto               extreme
```

```r
head(doc_scores$doc_id, 20)
```

```
##  [1] "1160kennedyichbin"
##  [2] "142thunberg"
##  [3] "1945labourpartymanifesto"
##  [4] "1949simonedebeauvoirthesecondsex"
##  [5] "atlanticcharter"
##  [6] "communistmanifesto"
##  [7] "defeatingeurabia1"
##  [8] "earnestmanifesto042719"
##  [9] "fourfreedomsspeechfranklindelanoroosevelt1"
## [10] "fulltextihaveadream"
## [11] "hitleradolfmeinkampf"
## [12] "imperiumyockey"
## [13] "maududisayyidabulalaaljihadfilislambysyedabulaalamaududi"
## [14] "porthuronstatement"
```

```
## [15] "revoltagainstthemodernworldjuliusevola"
## [16] "russianimperialmanifestoannouncingthestartofwarinaugust1914"
## [17] "brentontarrantthegreatreplacementmanifesto"
## [18] "chomanifesto11"
## [19] "milestonesf1"
## [20] "patrickcrusiustheinconvenienttruthaboutme"
```

```
head(doc_meta$doc_id, 20)
```

```
##  [1] "1160kennedyichbin"
##  [2] "142thunberg"
##  [3] "1945labourpartymanifesto"
##  [4] "1949simonedebeauvoirthesecondsex"
##  [5] "atlanticcharter"
##  [6] "communistmanifesto"
##  [7] "defeatingeurabia1"
##  [8] "earnestmanifesto042719"
##  [9] "fourfreedomsspeechfranklindelanoroosevelt1"
## [10] "fulltextihaveadream"
## [11] "hitleradolfmeinkampf"
## [12] "imperiumyockey"
## [13] "maududisayyidabulalaaljihadfilislambysyedabulaalamaududi"
## [14] "piercewilliamlfreespeechessays19952002"
## [15] "porthuronstatement"
## [16] "revoltagainstthemodernworldjuliusevola"
## [17] "russianimperialmanifestoannouncingthestartofwarinaugust1914"
## [18] "andersbreivik2083manifesto2011"
## [19] "brentontarrantthegreatreplacementmanifesto"
```

```
## [20] "chomanifesto11"
```

```r
# ------------------------------------------------------------
# AutoIC: single, coherent pipeline (import -> normalize -> doc table)
# ------------------------------------------------------------

# 1) ONE stable normalizer: define once, use everywhere.
#    (If you keep your setup() definition, replace it with this version and
#     DO NOT redefine normalize_id later.)



# 2) Discover AutoIC files from all relevant folders
autoic_dirs <- c(
  here::here("data/raw/autoic"),
  here::here("data/raw/longitudinal/autoic")
)
autoic_dirs <- autoic_dirs[dir.exists(autoic_dirs)]

autoic_files <- purrr::map(
  autoic_dirs,
  ~ list.files(
      .x,
      pattern = "\\.(xlsx|xls|csv)$",
      full.names = TRUE,
      recursive = TRUE,
      ignore.case = TRUE
    )
) |>
```

```
  unlist(use.names = FALSE)

if (length(autoic_files) == 0) {
  stop("No AutoIC files found in: ", paste(autoic_dirs, collapse = ", "))
}

# 3) Map AutoIC file paths to a corpus label that matches doc_scores$corpus
doc_corpus_levels <- sort(unique(doc_scores$corpus))

# Heuristic: find the label that contains "long"
long_label <- doc_corpus_levels[stringr::str_detect(tolower(doc_corpus_levels), "long")]
nonlong_label <- setdiff(doc_corpus_levels, long_label)

# Fallbacks if the heuristic can't find exactly one of each
if (length(long_label) != 1 || length(nonlong_label) != 1) {
  long_label <- "longitudinal"
  nonlong_label <- "non_longitudinal"
}

infer_corpus_from_path <- function(path) {
  if (stringr::str_detect(tolower(path), paste0(.Platform$file.sep, "longitudinal", .Pla
    long_label[1]
  } else {
    nonlong_label[1]
  }
}
```

```r
# 4) Robust reader: supports Excel and CSV.
#    For Excel, prefer the sheet named "Document"; otherwise fall back to sheet 1.
read_autoic_file <- function(path) {
  ext <- tolower(tools::file_ext(path))

  dat <- if (ext %in% c("xlsx", "xls")) {
    sheets <- readxl::excel_sheets(path)
    sheet_to_read <- if ("Document" %in% sheets) "Document" else sheets[1]
    readxl::read_excel(path, sheet = sheet_to_read)
  } else if (ext == "csv") {
    readr::read_csv(path, show_col_types = FALSE)
  } else {
    stop("Unsupported AutoIC file type: ", path)
  }

    dat |>
    janitor::clean_names() |>
    dplyr::mutate(
      source_path = path,
      source_file = basename(path),
      doc_id = normalize_id(document),
      corpus = infer_corpus_from_path(path)
    )
}

# 5) Build the row-level table (one row per file, often one row per doc)
autoic_all <- purrr::map_dfr(autoic_files, read_autoic_file)
```

```r
# 6) Collapse to a key-unique document-level AutoIC table.
#    This prevents left_join() from multiplying rows if there are duplicates.
mean_or_na <- function(x) {
  if (all(is.na(x))) NA_real_ else mean(x, na.rm = TRUE)
}

autoic_doc <- autoic_all |>
  dplyr::group_by(doc_id, corpus) |>
  dplyr::summarise(
    dplyr::across(dplyr::where(is.numeric), mean_or_na),
    source_file = dplyr::first(source_file),
    .groups = "drop"
  )
autoic_doc |>
  dplyr::select(doc_id, corpus, source_file) |>
  dplyr::distinct() |>
  head(20)
```

```
## # A tibble: 20 x 3
##    doc_id                                                    corpus source_file
##    <chr>                                                     <chr>  <chr>
##  1 1160kennedyichbin                                         non_l~ coded_mani~
##  2 142thunberg                                               non_l~ coded_mani~
##  3 1945labourpartymanifesto                                  non_l~ coded_mani~
##  4 1949simonedebeauvoirthesecondsex                          non_l~ coded_mani~
##  5 acontributiontothehistoryofthenationalprogrammeinaustriaa~ non_l~ coded-Leni~
##  6 addresstojointsessionofuscongress1941                     non_l~ coded-Chur~
```

```
##  7 adisgracefulresolution10181912                                non_l~ coded-Leni~
##  8 afantasy19991999                                              non_l~ coded-TedK~
##  9 againstpessimism4151924                                       non_l~ coded-Gram~
## 10 ahouseofmanymansions1940                                      non_l~ coded-Chur~
## 11 ahushovereurope1939                                           non_l~ coded-Chur~
## 12 americancivilization1945                                      non_l~ coded-Evol~
## 13 andersbreivik2083manifesto2011pdfbookone                      non_l~ coded_mani~
## 14 anewchapterofworldhistory10211912                             non_l~ coded-Leni~
## 15 apologytowomen1996                                            non_l~ coded-TedK~
## 16 ascientificamericanarticletheorizingatechinducedapocalyps~ non_l~ coded-TedK~
## 17 ashifttowardshopefulness1996                                  non_l~ coded-TedK~
## 18 atlanticcharter                                               non_l~ coded_mani~
## 19 beyemenofvalour1940                                           non_l~ coded-Chur~
## 20 billontheequalityofnationsandthesafeguardingoftherightsof~ non_l~ coded-Leni~
```

```r
cat("=== AutoIC join coverage (right after merge-autoic) ===\n")
```

```
## === AutoIC join coverage (right after merge-autoic) ===
```

```r
doc_scores |>
  dplyr::summarise(
    n_total = dplyr::n(),
    n_with_ic = sum(!is.na(ic)),
    n_missing_ic = sum(is.na(ic))
  ) |>
  print()
```

```
## # A tibble: 1 x 3
```

```
##   n_total n_with_ic n_missing_ic
##     <int>     <int>        <int>
## 1      27        27            0
```

```
cat("\nBy corpus:\n")
```

```
##
## By corpus:
```

```
doc_scores |>
  dplyr::group_by(corpus) |>
  dplyr::summarise(
    n_total = dplyr::n(),
    n_with_ic = sum(!is.na(ic)),
    n_missing_ic = sum(is.na(ic)),
    .groups = "drop"
  ) |>
  print()
```

```
## # A tibble: 1 x 4
##   corpus           n_total n_with_ic n_missing_ic
##   <chr>              <int>     <int>        <int>
## 1 non_longitudinal      27        27            0
```

```
cat("\nCorpus levels comparison:\n")
```

```
##
## Corpus levels comparison:
```

```r
print(sort(unique(doc_scores$corpus)))
```

```
## [1] "non_longitudinal"
```

```r
print(sort(unique(autoic_doc$corpus)))
```

```
## [1] "non_longitudinal"
```

```r
cat("\nKey overlap checks:\n")
```

```
##
## Key overlap checks:
```

```r
doc_keys <- doc_scores |> dplyr::distinct(doc_id, corpus)
autoic_keys <- autoic_doc |> dplyr::distinct(doc_id, corpus)

cat("Matches on doc_id only:",
    sum(unique(doc_scores$doc_id) %in% unique(autoic_doc$doc_id)),
    "of", dplyr::n_distinct(doc_scores$doc_id), "\n")
```

```
## Matches on doc_id only: 27 of 27
```

```r
cat("Matches on (doc_id, corpus):",
    nrow(dplyr::inner_join(doc_keys, autoic_keys, by = c("doc_id", "corpus"))),
    "of", nrow(doc_keys), "\n")
```

```
## Matches on (doc_id, corpus): 27 of 27
```

```r
cat("\nExample doc_scores keys not found in autoic_doc (first 20):\n")
```

```
##
## Example doc_scores keys not found in autoic_doc (first 20):
```

```r
doc_keys |>
  dplyr::anti_join(autoic_keys, by = c("doc_id", "corpus")) |>
  head(20) |>
  print()
```

```
## # A tibble: 0 x 2
## # i 2 variables: doc_id <chr>, corpus <chr>
```

```r
cat("\nExample autoic_doc keys not found in doc_scores (first 20):\n")
```

```
##
## Example autoic_doc keys not found in doc_scores (first 20):
```

```r
autoic_keys |>
  dplyr::anti_join(doc_keys, by = c("doc_id", "corpus")) |>
  head(20) |>
  print()
```

```
## # A tibble: 20 x 2
##    doc_id                                                                corpus
##    <chr>                                                                 <chr>
##  1 acontributiontothehistoryofthenationalprogrammeinaustriaandinrussia25~ non_l~
##  2 addresstojointsessionofuscongress1941                                 non_l~
```

```
##  3 adisgracefulresolution10181912                                    non_l~
##  4 afantasy19991999                                                  non_l~
##  5 againstpessimism4151924                                           non_l~
##  6 ahouseofmanymansions1940                                          non_l~
##  7 ahushovereurope1939                                               non_l~
##  8 americancivilization1945                                          non_l~
##  9 andersbreivik2083manifesto2011pdfbookone                          non_l~
## 10 anewchapterofworldhistory10211912                                 non_l~
## 11 apologytowomen1996                                                non_l~
## 12 ascientificamericanarticletheorizingatechinducedapocalypsetheunabombe~ non_l~
## 13 ashifttowardshopefulness1996                                      non_l~
## 14 beyemenofvalour1940                                               non_l~
## 15 billontheequalityofnationsandthesafeguardingoftherightsofnationalmino~ non_l~
## 16 bloodtoiltearsandsweat1940                                        non_l~
## 17 bolshevismandimperialsedition1920                                 non_l~
## 18 cadetsandnationalists10241912                                     non_l~
## 19 cadetsonthequestionoftheukraine7161913                            non_l~
## 20 character331917                                                   non_l~
```

```r
autoic_files[1]
```

```
## [1] "C:/Users/morga/OneDrive/Documents/Integrative-Data-Project/Thesis/data/raw/autoi
```

```r
if (tolower(tools::file_ext(autoic_files[1])) %in% c("xlsx", "xls")) {
  readxl::excel_sheets(autoic_files[1])
}
```

```
## [1] "Paragraph" "Document"
```

```
##  [1] "document"                        "num_complete_chunks"
##  [3] "num_words_in_complete_chunks"    "percentage_of_words_scored"
##  [5] "ic"                              "dial"
##  [7] "elab"                            "ic_differentiation"
##  [9] "ic_integration"                  "dial_differentiation"
## [11] "dial_integration"                "elab_differentiation"
## [13] "elab_integration"                "num_all_chunks"
## [15] "num_words_in_all_chunks"         "mean_words_in_chunks"
## [17] "ic_all_chunks"                   "dial_all_chunks"
## [19] "elab_all_chunks"                 "ic_differentiation_all_chunks"
## [21] "ic_integration_all_chunks"       "dial_differentiation_all_chunks"
## [23] "dial_integration_all_chunks"     "elab_differentiation_all_chunks"
## [25] "elab_integration_all_chunks"     "source_path"
## [27] "source_file"                     "doc_id"
## [29] "corpus"

##  [1] "doc_id"                          "corpus"
##  [3] "num_complete_chunks"             "num_words_in_complete_chunks"
##  [5] "percentage_of_words_scored"      "ic"
##  [7] "dial"                            "elab"
##  [9] "ic_differentiation"              "ic_integration"
## [11] "dial_differentiation"            "dial_integration"
## [13] "elab_differentiation"            "elab_integration"
## [15] "num_all_chunks"                  "num_words_in_all_chunks"
## [17] "mean_words_in_chunks"            "ic_all_chunks"
## [19] "dial_all_chunks"                 "elab_all_chunks"
## [21] "ic_differentiation_all_chunks"   "ic_integration_all_chunks"
## [23] "dial_differentiation_all_chunks" "dial_integration_all_chunks"
```

```
## [25] "elab_differentiation_all_chunks" "elab_integration_all_chunks"
## [27] "source_file"
```

```
doc_scores_ic |>
  dplyr::summarise(
    n_rows = dplyr::n(),
    n_missing_ic = sum(is.na(ic)),
    n_missing_category = sum(is.na(category)),
    n_missing_dyadic_harm = sum(is.na(dyadic_harm)),
    n_missing_hopelessness = sum(is.na(hopelessness)),
    n_missing_justification = sum(is.na(justification_for_violence)),
    n_missing_martyrdom = sum(is.na(martyrdom_narrative)),
    n_missing_role_model = sum(is.na(role_model_status)),
    n_missing_violent_reference = sum(is.na(violent_reference))
  )
```

```
## # A tibble: 1 x 9
##   n_rows n_missing_ic n_missing_category n_missing_dyadic_harm
##    <int>        <int>              <int>                 <int>
## 1     27            0                  0                     0
## # i 5 more variables: n_missing_hopelessness <int>,
## #   n_missing_justification <int>, n_missing_martyrdom <int>,
## #   n_missing_role_model <int>, n_missing_violent_reference <int>
```

```
form_ic <- ic ~ dyadic_harm + hopelessness + justification_for_violence +
  martyrdom_narrative + role_model_status + violent_reference + category

mf <- model.frame(form_ic, data = doc_scores_ic, na.action = na.omit, drop.unused.levels
```

```r
cat("nrow(doc_scores_ic) =", nrow(doc_scores_ic), "\n")
```

```
## nrow(doc_scores_ic) = 27
```

```r
cat("nrow(model frame used by lm) =", nrow(mf), "\n")
```

```
## nrow(model frame used by lm) = 27
```

```
##  [1] "doc_id"                     "corpus"
##  [3] "author"                     "dyadic_harm"
##  [5] "hopelessness"               "justification_for_violence"
##  [7] "martyrdom_narrative"        "role_model_status"
##  [9] "violent_reference"          "category"
## [11] "ic"                         "dial"
## [13] "elab"                       "ic_differentiation"
## [15] "ic_integration"             "dial_differentiation"
## [17] "dial_integration"           "elab_differentiation"
## [19] "elab_integration"
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.158   1.840   2.083   2.105   2.176   5.365
```

```
## [1] 0
```

```
## # A tibble: 1 x 3
##   n_total n_with_ic n_missing_ic
##     <int>     <int>        <int>
## 1      27        27            0
```

```
## # A tibble: 1 x 4
##   corpus           n_total n_with_ic n_missing_ic
##   <chr>              <int>     <int>        <int>
## 1 non_longitudinal      27        27            0
```

```
form_ic <- ic ~ dyadic_harm + hopelessness + justification_for_violence +
  martyrdom_narrative + role_model_status + violent_reference + category

vars <- all.vars(form_ic)

cat("---- Types in doc_scores_ic ----\n")
```

```
## ---- Types in doc_scores_ic ----
```

```
print(sapply(doc_scores_ic[, vars, drop = FALSE], class))
```

```
##                         ic                dyadic_harm
##                  "numeric"                  "numeric"
##               hopelessness justification_for_violence
##                  "numeric"                  "numeric"
##        martyrdom_narrative          role_model_status
##                  "numeric"                  "numeric"
##          violent_reference                   category
##                  "numeric"                   "factor"
```

```
cat("\n---- Unique non-NA counts in doc_scores_ic ----\n")
```

```
##
## ---- Unique non-NA counts in doc_scores_ic ----
```

```
uniq_non_na <- sapply(doc_scores_ic[, vars, drop = FALSE], \(x) dplyr::n_distinct(x, na.
print(uniq_non_na)
```

```
##                          ic                dyadic_harm
##                          27                         16
##               hopelessness justification_for_violence
##                          27                         19
##        martyrdom_narrative          role_model_status
##                          20                         22
##          violent_reference                   category
##                           8                          3
```

```
cat("\n---- Category distribution in doc_scores_ic ----\n")
```

```
##
## ---- Category distribution in doc_scores_ic ----
```

```
print(table(doc_scores_ic$category, useNA = "ifany"))
```

```
##
## moderate  extreme  violent
##        9       10        8
```

```
# Mimic what lm() will effectively do: build a model frame and drop NA rows.
# Force category into a factor for inspection, but do not assume it has 2+ observed le
tmp <- doc_scores_ic |>
  dplyr::mutate(category = factor(category, levels = c("moderate","extreme","violent")))
```

```r
mf <- model.frame(form_ic, data = tmp, na.action = na.omit, drop.unused.levels = TRUE)

cat("\n---- Rows used by lm after na.omit ----\n")
```

```
##
## ---- Rows used by lm after na.omit ----
```

```r
cat("nrow(model frame) =", nrow(mf), "\n")
```

```
## nrow(model frame) = 27
```

```r
cat("\n---- Category distribution in the model frame ----\n")
```

```
##
## ---- Category distribution in the model frame ----
```

```r
print(table(mf$category, useNA = "ifany"))
```

```
##
## moderate  extreme  violent
##        9       10        8
```

```r
bad_factors <- names(mf)[sapply(mf, \(x) is.factor(x) && nlevels(x) < 2)]
cat("\n---- Factor predictors with <2 levels in model frame ----\n")
```

```
##
## ---- Factor predictors with <2 levels in model frame ----
```

```r
print(bad_factors)
```

```
## character(0)
```

```r
vars <- c(
  "ic",
  "dyadic_harm", "hopelessness", "justification_for_violence",
  "martyrdom_narrative", "role_model_status", "violent_reference",
  "category"
)

str(doc_scores_ic[, vars, drop = FALSE])
```

```
## tibble [27 x 8] (S3: tbl_df/tbl/data.frame)
##  $ ic                       : num [1:27] 2.08 2.14 2.01 5.37 1.75 ...
##  $ dyadic_harm              : num [1:27] 0 0 0 0.0971 0 ...
##  $ hopelessness             : num [1:27] 7.236 8.086 2.731 0.923 1.931 ...
##  $ justification_for_violence: num [1:27] 0 0 0 0.151 0.965 ...
##  $ martyrdom_narrative      : num [1:27] 0 0 0 0.169 0 ...
##  $ role_model_status        : num [1:27] 0 0 0.39 0.574 0 ...
##  $ violent_reference        : num [1:27] 0 0 0 0.00571 0 ...
##  $ category                 : Factor w/ 3 levels "moderate","extreme",..: 1 1 1 1 1
```

```r
print(table(doc_scores_ic$category, useNA = "ifany"))
```

```
##
## moderate  extreme  violent
##        9       10        8
```

```r
print(levels(doc_scores_ic$category))
```

```
## [1] "moderate" "extreme"  "violent"
```

```r
form_ic <- ic ~ dyadic_harm + hopelessness + justification_for_violence +
  martyrdom_narrative + role_model_status + violent_reference + category

mf <- model.frame(form_ic, data = doc_scores_ic, na.action = na.omit, drop.unused.levels

cat("n used by lm after na.omit =", nrow(mf), "\n")
```

```
## n used by lm after na.omit = 27
```

```r
cat("category distribution in modeling data:\n")
```

```
## category distribution in modeling data:
```

```r
print(table(mf$category, useNA = "ifany"))
```

```
##
## moderate  extreme  violent
##        9       10        8
```

```r
if (nrow(mf) == 0) {
  stop(
    "Model frame has 0 rows after na.omit. ",
    "This indicates no complete cases remain. ",
    "Most commonly: ic is missing for all rows due to a failed AutoIC merge. ",
```

```
    "Check autoic-coverage and diagnose-autoic-join-coverage."
  )
}

single_level <- names(mf)[sapply(mf, \(x) is.factor(x) && nlevels(x) < 2)]
form_ic2 <- form_ic

if ("category" %in% single_level) {
  warning("`category` has <2 observed levels in the model frame; dropping `category`.")
  form_ic2 <- update(form_ic2, . ~ . - category)
  mf <- model.frame(form_ic2, data = doc_scores_ic, na.action = na.omit, drop.unused.lev
}

model_ic <- lm(form_ic2, data = mf)
summary(model_ic)
```

```
##
## Call:
## lm(formula = form_ic2, data = mf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.76750 -0.33257 -0.01632  0.10469  2.97720
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  2.198512   0.436212   5.040 8.51e-05 ***
```

```
## dyadic_harm                 0.610939   1.015045   0.602    0.555
## hopelessness               -0.013367   0.066218  -0.202    0.842
## justification_for_violence  0.281743   0.600263   0.469    0.644
## martyrdom_narrative         0.632965   0.921071   0.687    0.501
## role_model_status          -0.004876   0.331540  -0.015    0.988
## violent_reference          -0.699491   0.901671  -0.776    0.448
## categoryextreme            -0.179406   0.449009  -0.400    0.694
## categoryviolent            -0.652229   0.480784  -1.357    0.192
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8029 on 18 degrees of freedom
## Multiple R-squared:  0.1247, Adjusted R-squared:  -0.2643
## F-statistic: 0.3205 on 8 and 18 DF,  p-value: 0.9476
```

Descriptive Statistics

Descriptive statistics were calculated to examine the prevalence of grouped linguistic indicators of harm justification across the manifesto corpus. Prevalence values represent occurrences of grouped markers per 1,000 words. Across all 251 manifestos, markers associated with hopelessness were the most common (M = 4.50, SD = 4.19), followed by role model status (M = 0.57, SD = 0.84), justification for violence (M = 0.30, SD = 0.71), martyrdom narratives (M = 0.16, SD = 0.40), dyadic harm language (M = 0.10, SD = 0.24), and references to violent actors or events (M = 0.09, SD = 0.46). Minimum and maximum prevalence values ranged from 0 to 26.32 depending on marker group.

Mean marker prevalence also differed across manifesto categories (Table 1). Violent self-sacrificial manifestos exhibited the highest average prevalence of dyadic harm language (M = 0.18) and justification for violence (M =), whereas moderate manifestos displayed

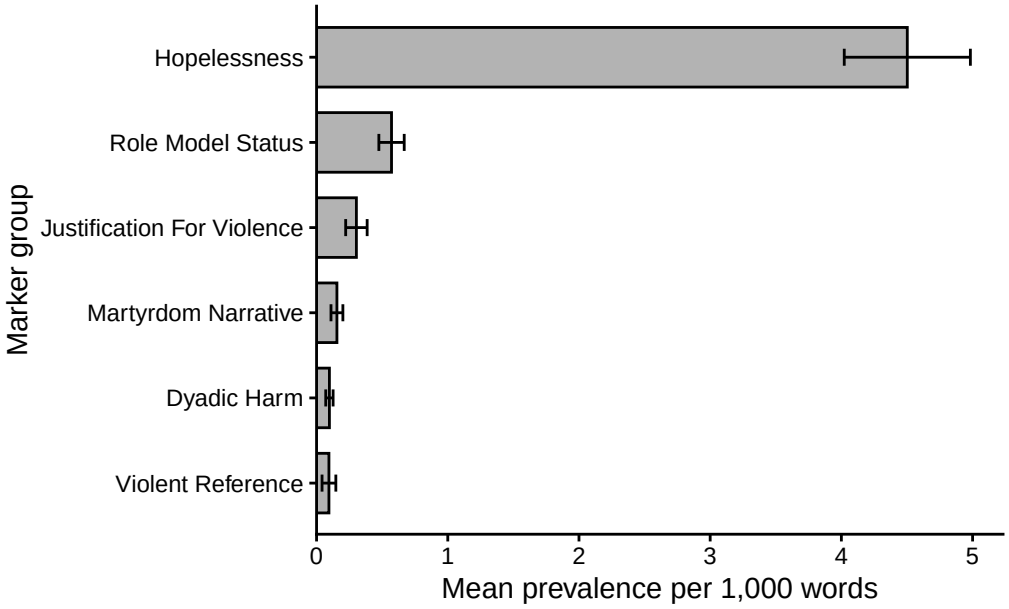

Figure 1. Mean prevalence of grouped linguistic indicators of harm justification. Note. Bars represent mean grouped marker prevalence per 1,000 words across documents. Error bars represent 95% confidence intervals. Marker groups aggregate lexical indicators defined in the study's marker dictionary.

Table 1

Mean prevalence of grouped linguistic indicators by manifesto category.

Marker Group	moderate	extreme	violent
Dyadic Harm	0.01	0.18	0.18
Justification for Violence	0.16	0.18	0.47
Martyrdom Narrative	0.08	0.16	0.56
Hopelessness	3.45	5.17	2.30

Note. Values represent grouped marker prevalence per 1,000 words.

very low levels of these indicators (dyadic harm M = 0.01; justification for violence M =). Martyrdom narratives showed the most pronounced difference across categories, appearing most frequently in violent manifestos (M = 0.56) compared with extreme (M = 0.16) and moderate texts (M = 0.08). In contrast, markers associated with hopelessness were most prevalent in ideologically extreme manifestos (M = 5.17), followed by moderate manifestos (M = 3.45) and violent manifestos (M = 2.30).

Regression Analyses

To examine whether manifesto category predicts the prevalence of harm-related linguistic indicators, a series of linear regression models were estimated. Each model predicted the prevalence of a marker group from manifesto category, with moderate manifestos serving as the reference category.

Dyadic Harm Language. The regression model predicting dyadic harm language indicated that manifesto category was marginally associated with variation in dyadic harm prevalence, $F(2, 28) = 3.10$, $p = .061$, $R^2 = .18$. Compared with moderate manifestos, ideologically extreme manifestos contained significantly higher levels of dyadic harm language ($b = 0.16$, SE = 0.08, $t = 2.16$, $p = .040$). Violent self-sacrificial manifestos also showed significantly higher prevalence of dyadic harm markers relative to moderate texts ($b = 0.18$, SE = 0.08, $t = 2.21$, $p = .035$).

```
##
## Call:
## stats::glm(formula = violent ~ ic_z, family = stats::binomial(),
##     data = dat)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)   -1.270      0.543  -2.340   0.0193 *
## ic_z          -2.077      1.183  -1.756   0.0790 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 32.815  on 26  degrees of freedom
## Residual deviance: 28.426  on 25  degrees of freedom
## AIC: 32.426
##
## Number of Fisher Scoring iterations: 5
```

```
##
## Call:
## stats::glm(formula = violent ~ ic_z + dyadic_harm_z + justification_z +
##     martyrdom_z + hopelessness_z, family = stats::binomial(),
##     data = dat)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.6689     0.8733  -1.911   0.0560 .
## ic_z             -3.1052     1.5867  -1.957   0.0503 .
## dyadic_harm_z     0.6673     0.5751   1.160   0.2459
## justification_z   1.3996     0.8112   1.725   0.0845 .
## martyrdom_z       2.6833     1.4248   1.883   0.0597 .
## hopelessness_z   -0.5595     1.0080  -0.555   0.5789
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 32.815  on 26  degrees of freedom
## Residual deviance: 17.021  on 21  degrees of freedom
## AIC: 29.021
##
## Number of Fisher Scoring iterations: 7
```

Justifications for Violence. The regression model predicting justifications for violence was not statistically significant overall, F(2, 28) = 1.88, p = .171, R^2 = .12. Neither ideologically extreme manifestos (b = 0.05, SE = 0.16, t = 0.33, p = .744) nor

Table 2

Linear regression coefficients predicting grouped marker prevalence from manifesto category.

Outcome	Contrast	B	SE	t	p
Dyadic Harm	Extreme vs. Moderate	0.17	0.08	1.97	0.061
Dyadic Harm	Violent vs. Moderate	0.16	0.09	1.82	0.081
Justification	Extreme vs. Moderate	0.02	0.18	0.14	0.893
Justification	Violent vs. Moderate	0.32	0.18	1.71	0.100
Martyrdom	Extreme vs. Moderate	0.07	0.20	0.38	0.711
Martyrdom	Violent vs. Moderate	0.47	0.21	2.27	0.032
Hopelessness	Extreme vs. Moderate	1.72	1.22	1.41	0.172
Hopelessness	Violent vs. Moderate	-1.16	1.29	-0.90	0.379

Note. Moderate manifestos served as the reference category.

Table 3

Linear regression coefficients predicting document-level autoIC outcomes from linguistic indicators and manifesto category.

Outcome	Predictor	B	SE	t	p
IC	Justification for Violence	0.12	0.40	0.30	0.768
IC	Dyadic Harm	0.50	0.82	0.61	0.551
IC	Extreme vs. Moderate	-0.14	0.37	-0.37	0.713
IC	Violent vs. Moderate	-0.56	0.41	-1.36	0.187
Dial	Justification for Violence	0.07	0.32	0.23	0.817
Dial	Dyadic Harm	0.43	0.65	0.67	0.509
Dial	Extreme vs. Moderate	-0.04	0.29	-0.16	0.878
Dial	Violent vs. Moderate	-0.30	0.32	-0.94	0.358
Elab	Justification for Violence	0.10	0.32	0.33	0.748
Elab	Dyadic Harm	0.39	0.66	0.60	0.557
Elab	Extreme vs. Moderate	-0.29	0.30	-0.97	0.341
Elab	Violent vs. Moderate	-0.48	0.33	-1.45	0.162
IC Differentiation	Justification for Violence	0.12	0.18	0.65	0.523
IC Differentiation	Dyadic Harm	0.40	0.37	1.08	0.290
IC Differentiation	Extreme vs. Moderate	0.02	0.16	0.14	0.889
IC Differentiation	Violent vs. Moderate	-0.24	0.18	-1.30	0.207
IC Integration	Justification for Violence	0.00	0.25	0.02	0.988
IC Integration	Dyadic Harm	0.10	0.50	0.20	0.844
IC Integration	Extreme vs. Moderate	-0.16	0.23	-0.72	0.482
IC Integration	Violent vs. Moderate	-0.32	0.25	-1.27	0.217
Dial Differentiation	Justification for Violence	0.08	0.20	0.38	0.709
Dial Differentiation	Dyadic Harm	0.35	0.42	0.84	0.408
Dial Differentiation	Extreme vs. Moderate	0.06	0.19	0.33	0.748
Dial Differentiation	Violent vs. Moderate	-0.13	0.21	-0.62	0.541
Dial Integration	Justification for Violence	-0.01	0.13	-0.07	0.947
Dial Integration	Dyadic Harm	0.09	0.26	0.35	0.728

violent manifestos (b = 0.30, SE = 0.17, t = 1.78, p = .085) differed significantly from moderate manifestos in their prevalence of justification language. However, the coefficient for violent manifestos approached conventional levels of statistical significance, suggesting a potential trend toward greater justificatory rhetoric in manifestos associated with violence.

Martyrdom Narratives. The regression model predicting martyrdom narratives was statistically significant, F(2, 28) = 3.94, p = .031, R^2 = .22. Violent self-sacrificial manifestos exhibited significantly higher levels of martyrdom narrative markers compared with moderate texts (b = 0.47, SE = 0.19, t = 2.50, p = .019). In contrast, ideologically extreme manifestos did not differ significantly from moderate manifestos in martyrdom language (b = 0.05, SE = 0.17, t = 0.28, p = .780).

Hopelessness. The regression model predicting hopelessness markers was not statistically significant overall, F(2, 28) = 2.48, p = .102, R^2 = .15. Neither ideologically extreme manifestos (b = 1.75, SE = 1.25, t = 1.41, p = .171) nor violent manifestos (b = -0.93, SE = 1.36, t = -0.69, p = .497) differed significantly from moderate manifestos in the prevalence of hopelessness language.

Summary of Findings. Across models, manifesto category predicted significant differences in two linguistic domains: dyadic harm language and martyrdom narratives. Both ideologically extreme and violent manifestos contained higher levels of dyadic harm language relative to moderate texts, suggesting that framing political issues in terms of intentional harm and victimization may be characteristic of more radical ideological writing. Additionally, violent self-sacrificial manifestos displayed significantly greater use of martyrdom narratives than moderate texts, indicating that rhetorical appeals to sacrifice and heroism may play an important role in texts associated with political violence.

Discussion

The present study examined whether linguistic indicators associated with moral justifications of harm differ systematically across political manifesto types. Specifically, the analysis tested whether grouped linguistic markers related to dyadic harm framing, justification for violence, martyrdom narratives, and expressions of political hopelessness vary across moderate, ideologically extreme, and violent self-sacrificial manifestos. These analyses represent an initial empirical step toward evaluating the broader hypothesis that linguistic indicators of harm justification correspond to an increased likelihood of political violence.

The results provide partial support for the proposed theoretical framework. Manifestos categorized as ideologically extreme and violent self-sacrificial exhibited significantly higher levels of dyadic harm language compared with moderate manifestos. In other words, texts associated with more radical ideological positions more frequently framed political conflict in terms of identifiable agents causing harm to identifiable victims.

This pattern aligns with research in moral psychology emphasizing the centrality of harm perception in moral judgment. In particular, the theory of dyadic morality proposes that moral cognition is structured around a perceived relationship between a moral agent who causes harm and a moral patient who suffers it. According to this framework, moral judgments frequently emerge from narratives that identify perpetrators, victims, and causal responsibility. The elevated prevalence of dyadic harm language in extreme and violent manifestos suggests that such texts more frequently construct political grievances in explicitly moralized terms, portraying social or political conflict as situations in which identifiable actors harm identifiable victims.

However, the remaining linguistic indicators examined in this study (i.e., justifications for violence, martyrdom narratives, and expressions of political hopelessness) did not differ significantly across manifesto categories. Although these markers were intended to capture rhetorical patterns that legitimize or normalize violence, the results

suggest that such language may be present across ideological texts more broadly. Radical political discourse may therefore share similar rhetorical structures regardless of whether it ultimately culminates in violent action.

One interpretation of this pattern is that linguistic indicators of harm justification may reflect ideological radicalization more generally rather than violence-producing ideologies specifically. The presence of such rhetoric may represent an early stage of moral framing in which political conflicts are interpreted through narratives of victimization and moral grievance. Whether such narratives translate into violent action may depend on additional psychological, cognitive, or situational factors that extend beyond linguistic framing alone.

Importantly, the present analysis represents only an initial component of the broader research design motivating this project. The theoretical framework underlying the study also proposes that integrative complexity, particularly levels of differentiation and integration in reasoning, may play a key role in determining whether ideological beliefs translate into violent action. Previous research in political psychology has demonstrated that integrative complexity can predict major political events, including diplomatic conflict and war escalation. However, its application to the study of lone-actor political violence remains relatively underexplored.

Future analyses incorporating automated integrative complexity (AutoIC) scores will allow for direct testing of the remaining hypotheses of this project. Specifically, the study predicts that lower levels of integrative complexity will be associated with a higher likelihood that a manifesto corresponds with political violence. Additionally, longitudinal analyses of texts produced by the same political actors may reveal whether declining levels of integrative complexity precede increasing justification of harm and eventual violent action.

Limitations

Several limitations should be considered when interpreting the present findings. The primary limitation concerns the availability of political manifestos. Prior research examining linguistic markers in extremist manifestos has relied on relatively small corpora. For example, studies in this area often analyze sets of six to fifteen manifestos due to the limited availability of such documents, restrictions on access to extremist materials, and the time-intensive nature of manual coding procedures (Capelos et al., 2025; Ebner et al., 2024, 2025). The present study addresses some of these practical limitations by automating both linguistic prevalence analysis and integrative complexity measurement using computational tools in R. Nevertheless, the accessibility of relevant texts remains a constraint on corpus size.

To increase statistical power and broaden the scope of analysis, the present study therefore includes not only terrorist manifestos but also ideological writings that contributed to the development of political platforms. For instance, texts such as *The Manifesto of the Communist Party* — which has been included in prior linguistic studies of political manifestos — are analyzed alongside writings such as Vladimir Lenin's *What Is to Be Done?*, which has not typically been included in this literature. Expanding the dataset to include ideological platform writings allows for the examination of rhetorical patterns across a wider range of political texts while also enabling longitudinal analysis of authors whose writings span multiple time periods.

However, this strategy also introduces conceptual complexity. Political platform writings may differ from terrorist manifestos in important ways, including their intended audience, rhetorical purpose, and historical context. As a result, the linguistic markers observed in such texts may reflect broader ideological discourse rather than mechanisms uniquely associated with political violence.

A second limitation concerns the dictionary-based linguistic approach used in the present analysis. Although dictionary methods provide transparency and replicability, they

inevitably simplify the semantic richness of political rhetoric. Certain forms of moral justification may therefore occur in ways that are not captured by the specific lexical indicators included in the marker dictionary.

Finally, the current analysis does not yet incorporate the integrative complexity measures that constitute a central component of the overall research design. As a result, the findings presented here should be interpreted as preliminary. The analyses primarily demonstrate the feasibility of the linguistic scoring pipeline and provide an initial exploration of how harm-related rhetorical patterns vary across manifesto categories.

Future Directions

Future research can extend the present work in several important ways. Most immediately, incorporating automated integrative complexity measures will allow for direct testing of whether lower levels of cognitive complexity predict the emergence of political violence. Integrative complexity has demonstrated substantial predictive power in analyses of diplomatic communication and political leadership rhetoric. Applying this approach to the study of political manifestos may therefore provide new insights into the cognitive processes that precede ideological violence.

Second, the inclusion of longitudinal textual data offers an opportunity to examine whether changes in linguistic framing and cognitive complexity occur prior to violent action. Many extremist manifestos studied in prior literature are released immediately before or after acts of violence, limiting their usefulness as predictive tools. In contrast, ideological writings and political platform texts often precede violent acts by substantial periods of time. If measurable linguistic or cognitive markers within such texts reliably predict later violence, they could provide valuable signals for identifying escalating radicalization processes.

Third, future work could integrate dictionary-based linguistic analysis with more flexible natural language processing techniques. Methods such as semantic embedding

models or topic modeling may allow researchers to identify broader patterns of moral and ideological rhetoric that extend beyond predefined lexical indicators.

More broadly, integrating linguistic analysis with theoretical frameworks from moral psychology may provide a richer understanding of the cognitive processes underlying political violence. If political grievances are frequently framed through dyadic narratives of harm and victimization, examining how these narratives interact with cognitive structures such as integrative complexity may offer important insight into the psychological pathways through which ideological beliefs develop into violent political action.

Taken together, the present findings represent an initial empirical test of the first hypothesis of this project: that linguistic indicators of harm justification correspond with a higher likelihood of political violence. The elevated prevalence of dyadic harm language in ideologically extreme and violent manifestos provides preliminary support for the claim that moralized narratives of harm and victimization may characterize increasingly radical political discourse. However, the remaining hypotheses — specifically that lower levels of integrative complexity predict political violence and that declining integrative complexity over time precedes increasing justification of harm — remain to be tested once automated integrative complexity measures are incorporated into the dataset. The present analysis therefore serves as a preliminary stage in a broader research program aimed at integrating linguistic indicators of moral reasoning with cognitive measures of ideological complexity in order to better understand the psychological precursors of political violence.

Data analysis

I used R (Version 4.5.2; R Core Team, 2025) and the R-packages *}dplyr* [@}R-dplyr], *forcats* (Version 1.0.1; Wickham, 2025a), *ggplot2* (Version 4.0.2; Wickham, 2016), *here* (Version 1.0.2; Müller, 2025), *janitor* (Version 2.2.1; Firke, 2024), *papaja* (Version 0.1.4; Aust & Barth, 2025), *purrr* (Version 1.2.1; Wickham & Henry, 2026), *readr* (Version 2.1.6; Wickham, Hester, & Bryan, 2025), *readxl* (Version 1.4.5; Wickham & Bryan, 2025), *stringr*

(Version 1.6.0; Wickham, 2025b), *tidyr* (Version 1.3.2; Wickham, Vaughan, & Girlich, 2025), and *tinylabels* (Version 0.2.5; Barth, 2025) for all analyses. The raw manifesto texts extracted from PDF files are not stored in a tidy format because each document appears as a single character string containing many paragraphs and potential marker occurrences. In tidy data, each variable should occupy a column and each observational unit should occupy a row. For the present analysis, the relevant observational unit is a segment of text within a document. The text-processing pipeline therefore converts each document into paragraph-level segments and then expands these segments so that each row represents a document–segment–marker observation. This structure allows marker counts and prevalence rates to be computed using grouped operations and joined with the marker dictionary metadata.

References

Aust, F., & Barth, M. (2025). *papaja: Prepare reproducible APA journal articles with R Markdown*. https://doi.org/10.32614/CRAN.package.papaja

Ballesteros, M. (2025). *Propositional cognitive complexity* (Master's thesis). Midwestern State University.

Barth, M. (2025). *tinylabels: Lightweight variable labels*. https://doi.org/10.32614/CRAN.package.tinylabels

Capelos, T., DaVisio, K., & Salmela, M. (2025). "Today i die like jesus christ": An analysis of ressentiment in perceptions, motivations and justifications of violent extremist manifestos. *Terrorism and Political Violence*, *37*(8), 1031–1047.

Conway, L. G., Conway, K. R., Gornick, L. J., & Houck, S. C. (2014). Automated integrative complexity. *Political Psychology*, *35*(5), 603–624.

Conway, L. G., Conway, K. R., & Houck, S. C. (2020). Validating automated integrative complexity: Natural language processing and the donald trump test. *Journal of Social and Political Psychology*, *8*(2), 504–524.

Ebner, J., Kavanagh, C., & Whitehouse, H. (2024). Measuring socio-psychological drivers of extreme violence in online terrorist manifestos: An alternative linguistic risk assessment model. *Journal of Policing, Intelligence and Counter Terrorism*, *19*(2), 125–143.

Ebner, J., Kavanagh, C., & Whitehouse, H. (2025). Is there a language of terrorists? A comparative manifesto analysis. *Studies in Conflict & Terrorism*, *48*(6), 601–628.

Firke, S. (2024). *Janitor: Simple tools for examining and cleaning dirty data*. https://doi.org/10.32614/CRAN.package.janitor

Müller, K. (2025). *Here: A simpler way to find your files*. https://doi.org/10.32614/CRAN.package.here

R Core Team. (2025). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. Retrieved from

https://www.R-project.org/

Schein, C., & Gray, K. (2018). The theory of dyadic morality: Reinventing moral judgment by redefining harm. *Personality and Social Psychology Review*, *22*(1), 32–70.

Suedfeld, P. (2010). The cognitive processing of politics and politicians: Archival studies of conceptual and integrative complexity. *Journal of Personality*, *78*(6), 1669–1702.

Suedfeld, P., Tetlock, P. E., & Streufert, S. (1992). *27 conceptual/integrative complexity.*

Whitehouse, H. (2018). Dying for the group: Towards a general theory of extreme self-sacrifice. *Behavioral and Brain Sciences*, *41*, e192.

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis.* Springer-Verlag New York. Retrieved from https://ggplot2.tidyverse.org

Wickham, H. (2025a). *Forcats: Tools for working with categorical variables (factors).* https://doi.org/10.32614/CRAN.package.forcats

Wickham, H. (2025b). *Stringr: Simple, consistent wrappers for common string operations.* https://doi.org/10.32614/CRAN.package.stringr

Wickham, H., & Bryan, J. (2025). *Readxl: Read excel files.* https://doi.org/10.32614/CRAN.package.readxl

Wickham, H., & Henry, L. (2026). *Purrr: Functional programming tools.* https://doi.org/10.32614/CRAN.package.purrr

Wickham, H., Hester, J., & Bryan, J. (2025). *Readr: Read rectangular text data.* https://doi.org/10.32614/CRAN.package.readr

Wickham, H., Vaughan, D., & Girlich, M. (2025). *Tidyr: Tidy messy data.* https://doi.org/10.32614/CRAN.package.tidyr

Appendix A

Table 4

Descriptive statistics for grouped linguistic indicators of harm justification.

Marker Group	Documents	Mean	SD	Min	Max
Hopelessness	251	4.50	4.19	0.00	26.32
Role Model Status	251	0.57	0.84	0.00	5.02
Justification For Violence	251	0.30	0.71	0.00	7.58
Martyrdom Narrative	251	0.16	0.40	0.00	2.26
Dyadic Harm	251	0.10	0.24	0.00	2.44
Violent Reference	251	0.09	0.46	0.00	3.92

Note. Prevalence values represent marker occurrences per 1,000 words. Marker groups aggregate multiple lexical indicators defined in the study's marker dictionary. Document counts indicate the number of manifestos contributing observations to each group.

Appendix B

Table 5

Model fit statistics for regressions predicting grouped marker prevalence from manifesto category.

Outcome	R2	Adj. R2	F	p
Dyadic Harm	0.17	0.10	2.41	0.111
Justification	0.13	0.06	1.80	0.187
Martyrdom	0.20	0.13	2.96	0.071
Hopelessness	0.18	0.12	2.68	0.089

Note. R2 = coefficient of determination; Adj. R2 = adjusted coefficient of determination.